

Homework Assignment 7

Connor Mooneyhan

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I waited too long to start thinking about these. I'll be back on track next week.

1. (2E.3) Give an example of a sequence of continuous functions $f_1, f_2, \dots$ from $[0, 1]$ to $\mathbb{R}$ that converges pointwise to a function $f : [0, 1] \rightarrow \mathbb{R}$ that is not a bounded function.
2. (2E.6) Suppose (X, S, μ) is a measure space with $\mu(X) < \infty$. Suppose $f_1, f_2, \dots$ is a sequence of S -measurable functions from X to $\mathbb{R}$ such that $\lim_{k \rightarrow \infty} f_k(x) = \infty$ for each $x \in X$. Prove that for every $\varepsilon > 0$, there exists a set $E \in S$ such that $\mu(X \setminus E) < \varepsilon$ and $f_1, f_2, \dots$ converges uniformly to ∞ on E (meaning that for every $t > 0$, there exists $n \in \mathbb{Z}^+$ such that $f_k(x) > t$ for all integers $k \geq n$ and all $x \in E$.)
3. (3A.3) Suppose (X, S, μ) is a measure space and $f : X \rightarrow [0, \infty]$ is an S -measurable function. Prove that

$$\int f \, d\mu > 0 \text{ if and only if } \mu(\{x \in X : f(x) > 0\}) > 0.$$

4. (3A.4) Give an example of a Borel measurable function $f : [0, 1] \rightarrow (0, \infty)$ such that $L(f, [0, 1]) = 0$.
5. (3A.5) Verify the assertion that integration with respect to counting measure is summation.